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Abstract

Object Detection has very important role in video analysis and image processing, in recent years it has attracted much attention in research field. In this dissertation entitled “Moving Object Detection based on Background Subtraction using Energy Function” an Object Detection method has been proposed that based on segmentation of moving object elements from static background using Energy function. Firstly, in each frame key points (Interesting Pixels) are detected using ORB and KAZE algorithm instead of processing every pixels of a frame. Then Energy Function is applied to subtract moving object’s key points from static key points by calculating entropy of each pixel in each frame. Here entropy is calculated using Gaussian Distribution / Normal Distribution. Finally, those key points are grouped into different objects using Morphological Operations and Blob Detection method.

Chapter 1:

Introduction

One of the most important senses of humans and many other animals, is visual perception. Whenever we want to interact with our environment we mostly rely on vision. As when we want to pick up an object, we walk or drive through our environment and avoid clash with something on our way or when we are able to recognize our friends so easily. For all these type of tasks, object detection and localization is essential. In order to pick up a book, first thing we need to determine within our visual sight among so many objects which one is the book we are fetching and where it is located then we have to move our hands to grasp it.

Of course, we do not actively think about all these basic steps but this is a huge challenge for artificial system like robots what our brain doing effortlessly. In Computer Vision there are several subtasks of the task to understand and to solve different problems of a visual content like image or video or simply to achieve at least a part of human visual perception abilities. Object detection is one of those subtasks which helps in larger areas of Computer Vision.

1.1 Object detection

Computer vision aims at using computers to serve similar functions of human vision, in recent years it has attracted much attention in research field. It has a wide range of subtopics such as object detection, object recognition, image classification, image retrieval, and many others. Among these subtopics, object detection, as the task of locating and detecting objects in images and videos, is challenging and one of the most important tasks in computer vision.

Object detection is a computer vision technique that allows us to identify and locate objects in an image or video. With this kind of identification and localization, object detection can be used to count objects in a scene and determine and track their precise locations, all while accurately boxing them.

The goal of object detection[1] is to detect all objects in an image. Typically only a small number of objects are present in the image, but there is a very large number of possible locations and scales at which they can occur and that need to somehow be explored. Each detection is reported with some form of pose information. This could be as simple as the

location of the object, a location and scale, or the extent of the object defined in terms of a bounding box. In other situations the pose information is more detailed and contains the parameters of a linear or non-linear transformation. For example a face detector may compute the locations of the eyes, nose and mouth, in addition to the bounding box of the face. An example of object detection that specifies the locations of certain parts is shown in Figure 1.1.

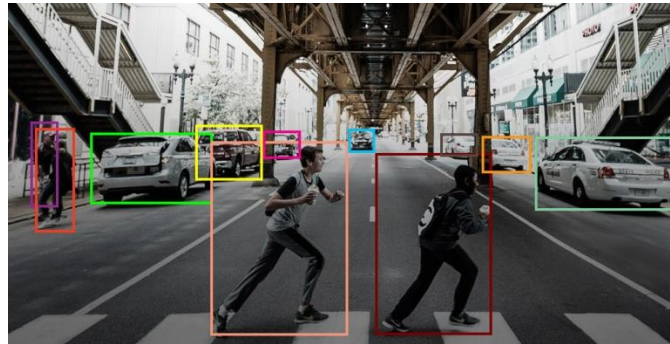


Figure 1.1 : An example of Object Detection

1.2 Moving object detection and it's application

Moving object detection is a computer technology related to computer vision, image processing that deals with detecting moving objects of a certain category (such as human, cars etc) in digital image or video. Researched domains include vehicle detection, human detection. Moving object detection has many applications in the domain of computer vision, including image retrieval and video surveillance.

Moving object detection[2] is the detection process of finding the location of moving objects in images or videos. Moving object detection involves locating objects in the frame of a video sequence. It's not easy task to do in video as the video may be taken with static camera or may be with dynamic / moving camera. In the first case detection is a bit easier as background of the video is static. But in the second case the background also dynamic. So for the second scenario both objects and background are moving that's why detection of only moving foreground or moving objects is a bit tough. In moving object detection various background subtraction techniques available. Background subtraction involves the absolute difference between the current image and the reference updated background over a period of time. A good background subtraction should be able to overcome the problem of varying illumination condition, background shadows, camouflage, bootstrapping and at the same time motion segmentation of foreground

object should be done at the real time[3]. An example of moving object detection that specifies detection of moving cars on the road where everything is static apart from those cars in Figure 1.2.



Figure 1.2 : Moving object detection

The moving object tracking in video pictures has attracted a great deal of interest in computer vision. Object detection is the first step in surveillance systems, navigation systems and object recognition [4]. The CCTV is one of the main reasons for the growing interest and use of video in security systems. Moving object detection in a video stream is an essential step in video surveillance applications. In some algorithms, the moving objects may become part of the scene when they come to a stop. Also the scene maybe affected by changes in the light, leaves swaying, cameras shaking, etc. Many algorithms of moving object detection has been proposed in recent years. These involve background subtraction, optical flow, temporal difference and many other algorithms for detecting moving objects. From these, the most widely used algorithm is background subtraction, which has many algorithms such as frame difference, approximate median, Gaussian mixture.

Moving Object detection has huge significance in real time environment. As it has several important applications such as it is used to provide better sense of security using visual information. It's also used to give security and surveillance to recognize people, to analyze shopping behavior of customers in retail space, video abstraction to obtain automatic annotation of videos, to generate object based summaries, traffic management to analyze flow, to detect accidents, video editing to eliminate human operator interaction.

Chapter 2:

Literature Survey

2.1 Types of detection methods

The detection of an object in video sequence plays a significant role in many applications. Especially the video surveillance has long been in the use for monitoring security sensitive areas for example banks, departmental stores, traffic monitoring on highway, public places which are crowded. Moving object detection methods are used to detect a moving object like human, vehicles etc. Various methods that are used to detect the moving object, are shown in Figure 2.1.

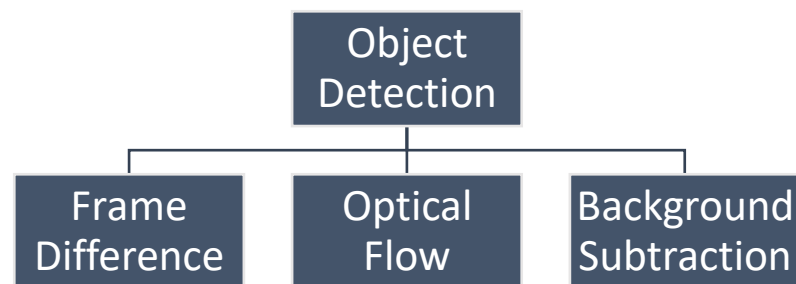


Figure 2.1 : Types of object detection method

2.1.1 Frame difference

This method uses the two adjacent frames based on time series image to subtract and gets difference images, its working is very similar to background subtraction method [5]. After the subtraction of image it gives moving target information through the threshold value. This method is simple and easy to implement. This is very adaptive to dynamic scene changes, however, it generally fails in detecting whole relevant pixels of some types of moving objects. Additional methods need to be adopted in order to detect stopped objects for success of higher levels are computationally complex and cannot be used real-time without specialized hardware.

C. Zhan et al.(2007) [6] proposed a method which is an improved algorithm based on frame difference and edge detection is presented for moving object detection. It detects the edges of each two continuous frames by Canny detector and gets the difference between the two edge images. And then, it divides the edge difference image into several small blocks and decides if they are moving areas by comparing the number of non-zero pixels to a threshold. At last, it does the block-connected component labeling to get the smallest rectangle that contains the moving object. It has a high recognition rate and a high detection speed, which has a broad market prospect.

I. Kartika et al.(2011) [7] proposed a method using one of the traditional background subtraction techniques which is frame differencing (FD) is conducted to detect the moving object in outdoor environment. The result of moving object detection using FD is not perfect that enable this research to experimental two existing post-processing techniques which are adaptive threshold and shadow detection in HSV color space for outdoor environment. Experimental result showed that FD with post-processing techniques are able to detect good moving object in outdoor environment with no shadow based on accuracy improvement data.

N. Srivastav et al.(2017) [8] proposed a method, a video sequence containing a large number of images is used to extract motion information. Two frame differencing is very easy but there is problem of holes. Three frame differencing and background subtraction have solved the problem of holes of two frames till a limit. Background subtraction is used for stable background video but Dynamic Background subtraction is capable to detect object in video with gradual background changes.

2.1.2 Optical flow

The optical flow method uses the motion target of the vector characteristics which changed with time to detect moving area in video [5]. It gives better performance under moving camera, but this algorithm is very complex and complicated computation and also it needs special hardware support, so it is difficult to meet the requirements of real-time video processing.

H. Zhang (2010) [9] proposed a multiple object detection and tracking algorithm based on optical flow in polar-log images. Optical flow computation is only used in moving area. Firstly, the moving edge is extracted in polar-log coordinate. Secondly, the generalized dynamic image model (GDIM) based method is used and the gradient operator in polar-log coordinate is employed to compute the optical flow directly in every moving area and the object tracking is accomplished. This method has there advantages. One is that the image size reduced and the computing time of optical flow decreased in polar-log coordinate. The other is that the optical flow calculation result

is accurate because the GDIM base method is used and the gradient operator in polar-log coordinate is employed. The third is that excessively smooth can be resolved by optical flow computation only used in moving area. And this method can be used for multiple objects tracking and real time object tracking.

K. Kale et al. (2015) [10] proposed a new object detection and tracking algorithm which makes use of optical flow in conjunction with motion vector estimation for object detection and tracking in a sequence of frames. The optical flow gives valuable information about the object movement even if no quantitative parameters are computed. The motion vector estimation technique can provide an estimation of object position from consecutive frames which increases the accuracy of this algorithm and helps to provide robust result irrespective of image blur and cluttered background. The use of median filter with this algorithm makes it more robust in the presence of noise. The developed algorithm is applied to wide range of standard and real time datasets with different illumination (indoor and outdoor), object speed etc.

X. Han et al. (2015) [11] proposed a moving detection algorithm that combine with optical flow method and three frame difference method which be used OTSU to process. Choose the typical Harris corner of extracting area to calculate, it can effectively reduce the complexity of the algorithm. Since moving target area detection was not complete, the three-frame difference image which be processed by OTSU binary image was introduced as a supplement of simplify the optical flow method. By this proposed method the optical flow method is satisfied with the real-time request and more complicated areas of the moving object are detected.

2.1.3 Background subtraction

It is particularly a commonly used technique for object detection in static images [5,12,13]. It will detect moving regions by subtracting the current image pixel-by-pixel from a reference background image that is created by averaging images over time in an initialization period. The basic idea of background subtraction method is to initialize a background firstly, then by subtracting current frame in which the moving object present that current frame is subtracted with background frame to detect moving object. This method is simple and easy to realize, and accurately extracts the characteristics of target data, but it is sensitive to the change of external environment, so it is applicable to the condition that the background is known.

S. S. Mohamed et al. (2010) [14] suggested to use background subtraction technique using Mixture of Gaussian (MoG) method is conducted for detection of moving object at outdoor environment. Focus is specified at the five parameters of MoG namely background component weight threshold (T_s), standard deviation scaling factor (D), user-define learning rate (α), Total number of Gaussian components (K) and

Maximum number of components M in the background model (M) to give significant impact in producing the optimize background subtraction process. Experimental results showed that by varying each of the parameter can produce acceptable results that enable us to propose suitable parameter range of each parameter for detection of moving object in an outdoor environment.

D. M. Rashed et al. (2013) [15] introduce a new algorithm based on adaptive background subtraction that has reduced computation complexity, hence, lower processing time while maintaining a competitive performance in terms of the recall and precision parameters. The proposed algorithm construct background model and compares its pixels with the current images to identify foreground/background pixels and minimizes the number of updated pixels in background model to reduce the processing time. The processing time is decreased by 74% compared to the SOBS algorithm. In addition, the proposed algorithm has competitive performance with respect to state-of-the-art algorithms in moving object detection.

J. Lee et al. (2014) [16] proposed a method that detects moving objects in depth image sequences using background images and motion depth (MD) values. The background image represents the camera view with no moving objects and the MDs are the depth values of moving objects. Foreground regions can be easily detected by background subtraction; however, the foreground regions have some noise and do not contain a regional diffusion of depth values. Therefore, we apply both background subtraction and motion depth detection to detect moving objects.

R. D. Sharma et al. (2017) [17] proposed in this paper, background is updated with averaging of frame $t-1$, frame $t+1$ and previous updated background because dynamic background detection technique is much better than static background technique for video with background change. This updated background is subtracted from frame t for foreground detection and merged with three frame subtraction. So there is scope of work such that holes problem should be reduced more and object should be detected better in dynamic changes in background. In this work, the proposed technique is able to reduce the holes problem in dynamic background updating video. This technique is extract foreground better than existing static and dynamic background.

W. Wan et al. (2019) [18] investigated The problem of moving object detection based on high-speed video sequence images is in this paper. First, the high-speed video tracking system is introduced. Second, an improved hybrid detection method combined background subtraction with frame difference is proposed, and the moving object is post-processed by using the threshold area de-noising and morphology filter. Finally, the experiment results show that the proposed method detects the moving object from the high-speed video sequence images precisely and effectively.

Z. Charouh et al. (2019) [19] proposes a method to adjust in real-time the morphological parameters to the illumination changes in the scene. The method is

based on machine learning. The features used in the machine learning models are first, second, third and fourth-order statistics of the grayscale images, and the outputs are the appropriate morphological parameters. The resulting background subtraction-based object detection is shown to be robust to illumination changes, and to significantly outperform the conventional approach. Further, artificial neural network (ANN) is shown to provide better performance than Naive Bayes and K-Nearest Neighbours models.

Chapter 3:

Proposed method – Moving Object Detection based on Background Subtraction using Energy function

3.1 System overview

Detection of dynamic or moving object is an important task in the area of computer vision application. In this object detection method objects in consecutive frames of a video file are located or detected. To detect the moving objects in a video modified frame difference algorithm along with subtraction of the dynamic foreground from the static background by using energy function are used and then further thresholds are applied. For grouping those key points into objects morphological operations are blob detection algorithm are used. Workflow of this method is shown is Figure 3.1 below.

There are three basic or main steps:

- Detection of interesting pixels as key points from each frame.
- Calculation of energy in each frame in the form of entropy using modified energy function and detection of moving objects key points by a threshold.
- For grouping those key points Morphological operations (Erosion and Dilation) and Blob detection method are applied and put bounding box around the objects.

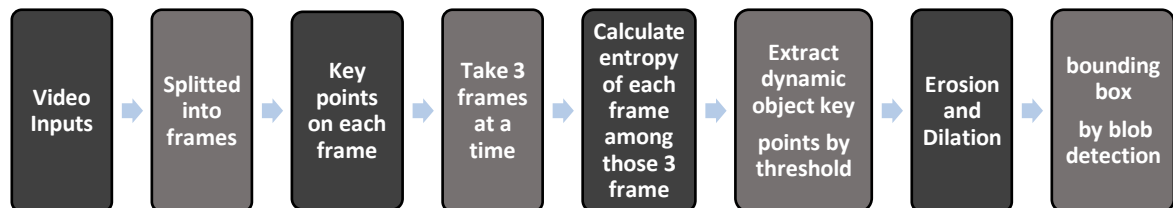


Figure 3.1 : The workflow of moving object detection

3.1.1 Input

Video is a collection of multiple images those are constantly displayed to create motion effects. Thus, object detection in video files is actually detection of objects in image file (cause as mentioned earlier that video is made of multiple images). We need to perform the conversion of video into frames to determine where the areas of moving objects are, and where the area of the background is. Usually one second of video has 24 (or maybe more) still images and 24 or more photos. So the video which is taken as input is converted into frames with rate of 4 frame per second .

Let after conversion the video sequence V has n frames denoted as, where one frame denoted as $f(x, y)$.

A frame is nothing but an image. An image is basically a 2D array where x is number of rows and y is number of columns.

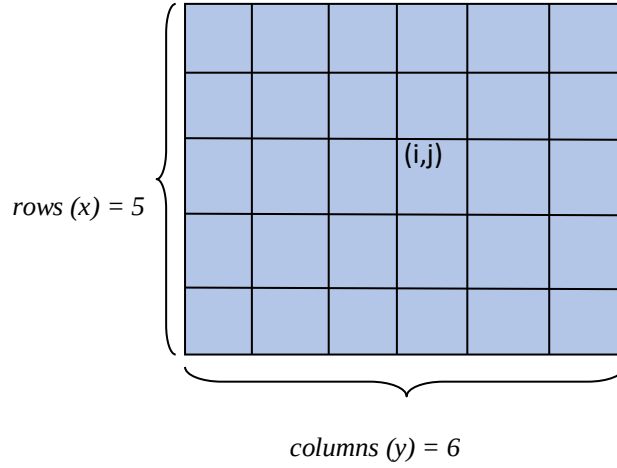


figure 3.2 : A 2D image representation

A video consists of images which has same dimension. That means number of rows x and number of columns y is same.

$$V = \{ f_1(x, y), f_2(x, y), f_3(x, y), \dots, f_n(x, y) \} \quad \dots(3.1)$$

$$\text{or,} \quad V = \sum_{i=1}^n f_i(x, y) \quad \dots(3.2)$$

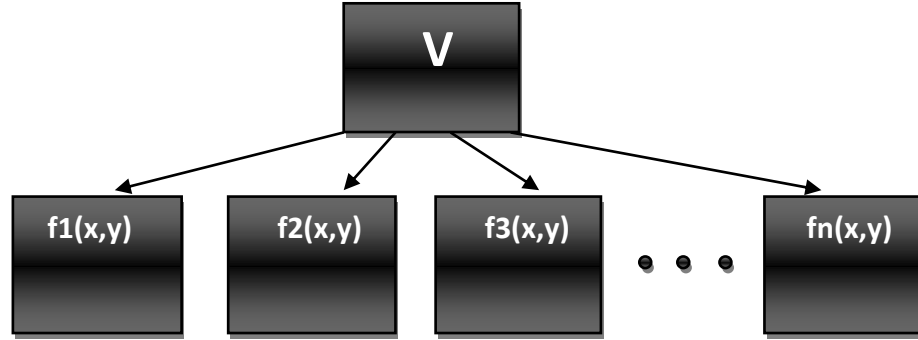


figure 3.3 : Video V is converted into n consecutive frames

3.1.2 Key points detection

A video contains many number of frames and in each frame there is so many number of pixels. Suppose if a single frame have 100 row and 100 columns so it has 10000 pixels and a video has 10 sec video with 4fps. So total pixels for that will be 4,00,000 pixels. Working with these huge number of pixels will increase the computation time. To avoid this unnecessary computation important pixels or key points has been detected with feature detector algorithm.

A feature-detector is an algorithm that detects feature-points (also called interest-points or key-points) in an image [20]. During this step, important key points are searched in an image. An important or interesting key point is a pixel in the image that differs from its immediate neighborhood in terms of intensity, color, and texture. ORB [25] and AKAZE [26] feature detection algorithms are used in this detection algorithm.

ORB (Oriented FAST and Rotated BRIEF) algorithm is a blend of modified FAST (Features from Accelerated Segment Test) [23] detection and direction-normalized BRIEF (Binary Robust Independent Elementary Features)[24] description methods. Here only FAST [22] algorithm is used to detect important pixels of each frame.

Let's take the i^{th} frame $f_i(x,y)$ and from that the m^{th} pixel $p_i(x_m, y_m)$ is selected which is to be checked if it is an interest point or not. Let its intensity be $I_i(x_m, y_m)$ this algorithm described in the figure 3.4 below.

Around the selected pixel a circle of 16 pixels is considered.

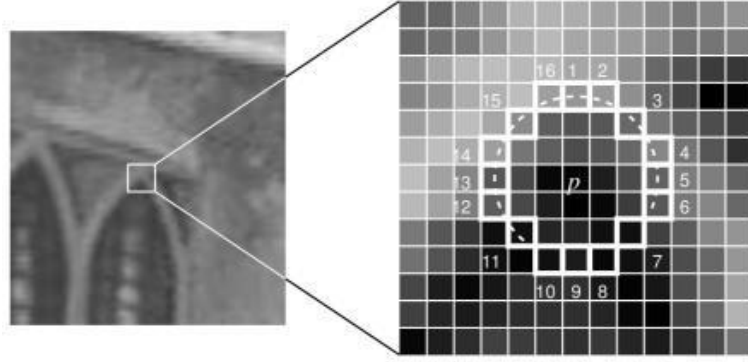


Figure 3.4 : Important pixel selection with FAST algorithm

Now only the four pixels at 1, 9, 5 and 13 (First 1 and 9 are tested if they are too brighter or darker. If so, then checks 5 and 13) and an appropriate threshold value t is selected. Here value of t is taken as 50.

If $p_i(x_m, y_m)$ is a key point, Then at least three of these must all be brighter than $I_i(x_m, y_m) + t$ or darker than $I_i(x_m, y_m) - t$. If neither of these is the case, then $p_i(x_m, y_m)$ cannot be a important pixel. This important pixel selection described in the figure 3.5 below:

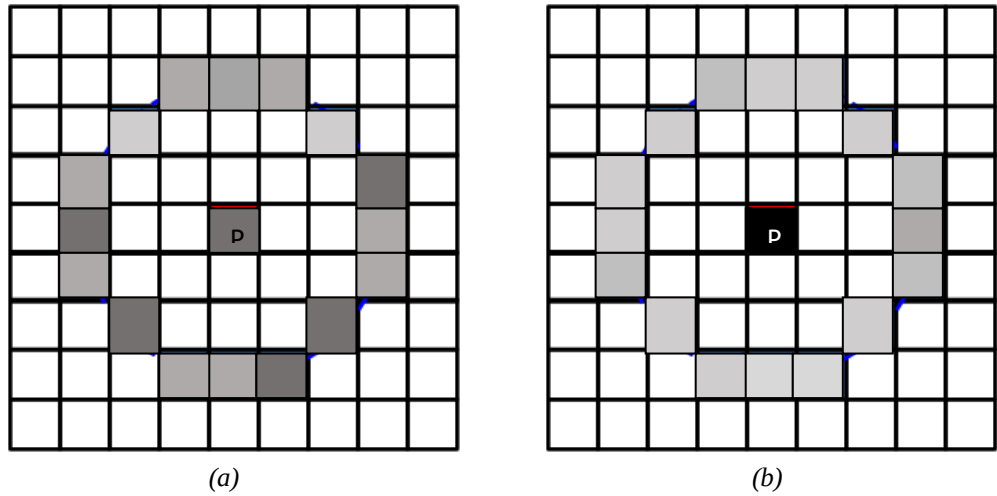


Figure 3.5 : (a) It is not a key point as $I_i(x_m, y_m)$ has not much difference from the intensities of the surrounded pixel, (b) it is a key point as its intensity is much lower than the surrounded pixels.

Likewise all the key points k are extracted from that frame $f_i(x, y)$ through ORB method stored in kp_i^{orb}

$$kp_i^{orb} = \{k_1, k_2, k_3, \dots\} \quad \dots(3.3)$$

ORB method are not invariant of noise so to get more accurate key points AKAZE method[21] is also used. This algorithm uses nonlinear diffusion filtering instead of Gaussian filter to blur the image. Nonlinear diffusion does not create or destroy actual object structure.

Following is partial-differential-equation (PDE) based nonlinear diffusion filter [27] and lets take the i^{th} frame $f_i(x, y)$ and m^{th} pixel:

$$P_i(x_m, y_m) = \text{div}[c(|\nabla f_i(x_m, y_m)|) \cdot \nabla f_i(x_m, y_m)] \quad \dots(3.4)$$

Where $c(\cdot)$ is the diffusion coefficient function based on the image gradient $\nabla f_i(x_m, y_m)$. Diffusion coefficient functions of the form:

$$c(x) = \frac{1}{1+(x/k)^2} \quad \dots(3.5)$$

Where k is the noise threshold that controls the amount of diffusion to be applied in the gradient direction.

And the gradient $\nabla f_i(x_m, y_m)$ is :

$$\nabla f_i(x_m, y_m) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f_i(x_m, y_m)}{\partial x} \\ \frac{\partial f_i(x_m, y_m)}{\partial y} \end{bmatrix} \quad \dots(3.6)$$

Where, g_x is the derivative with respect to x (gradient in the x direction) and g_y is the derivative with respect to y (gradient in the y direction).

AKAZE uses the determinant of the Hessian (DoH) matrix[28]. DoH images are computed in each image. The Hessian matrix H is of second-order partial derivatives of P :

$$H = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{xy} & I_{yy} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 P_i(x_m, y_m)}{\partial x^2} & \frac{\partial^2 P_i(x_m, y_m)}{\partial x \partial y} \\ \frac{\partial^2 P_i(x_m, y_m)}{\partial y \partial x} & \frac{\partial^2 P_i(x_m, y_m)}{\partial y^2} \end{bmatrix} \quad \dots(3.7)$$

As $\frac{\partial^2 P_i(x_m, y_m)}{\partial x \partial y}$, $\frac{\partial^2 P_i(x_m, y_m)}{\partial y \partial x}$ is same so both are denoted as I_{xy} . So the determinant of the Hessian matrix (H) is DoH as follow:

$$\text{DoH} = \det(H) = (I_{xx}I_{yy} - I_{xy}^2) \quad \dots(3.8)$$

The key points (or features) are extracted by comparing the pixels in DoH images with a surrounding window of size 3×3 . If the value of this pixel is greater than the

other 8 surrounding pixels and a predefined threshold then it is considered as a key point.

Likewise all the key points k are extracted from that frame $f_i(x, y)$ by AKAZE method kept in kp_i^{akaze} .

$$kp_i^{akaze} = \{k_1, k_2, k_3, \dots\} \quad \dots(3.9)$$

Key points detected with both ORB and AKAZE algorithms are combined for each frame. Before applying the methods all frames are converted from BGR to RGB color scheme.

So the final key points of the i^{th} frame $f_i(x, y)$ would store in kp_i^{total} and this union is described in Figure 3.6 below,

$$kp_i^{total} = (kp_i^{orb} \cup kp_i^{akaze}) \quad \dots(3.10)$$

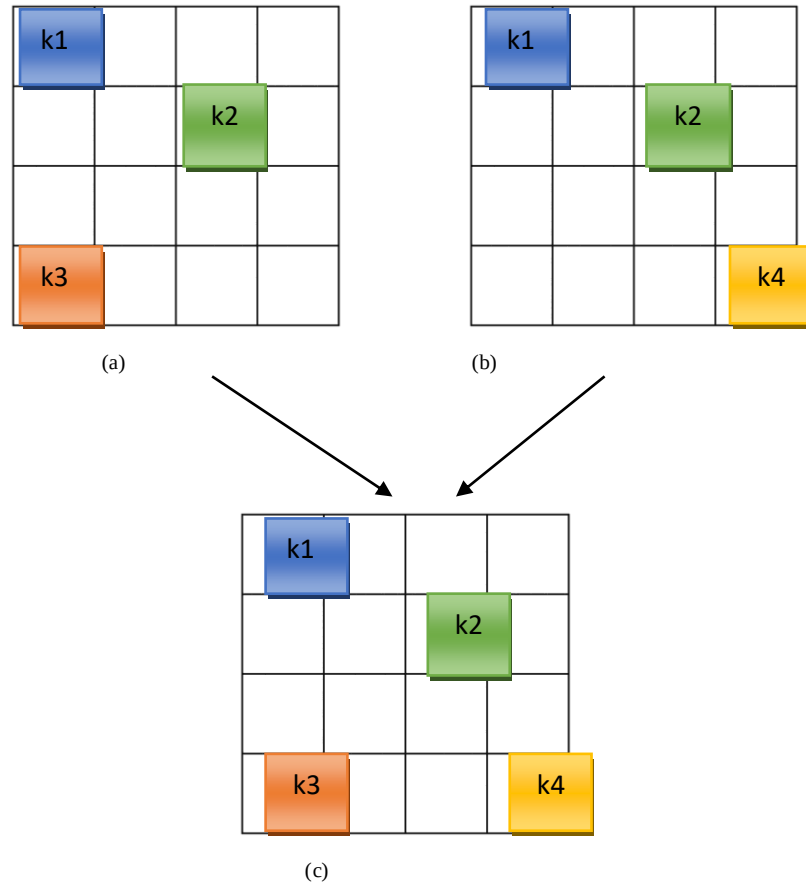


Figure 3.6: (a) kp_i^{orb} where k_1, k_2, k_3 are key points extracted by ORB method, (a) kp_i^{akaze} where k_1, k_2, k_4 are key points extracted by AKAZE method, (c) kp_i^{total} is the union of kp_i^{orb} and kp_i^{akaze} where key points are k_1, k_2, k_3, k_4 .

Examples of ORB and AKAZE key points are given in (Figure 3.7, 3.8) where in Figure 3.7 ORB is less dense but AKAZE are more dense but in Figure 3.8 is opposite. This is why to get more number important pixels around the objects ORB and AKAZE key points are combined.

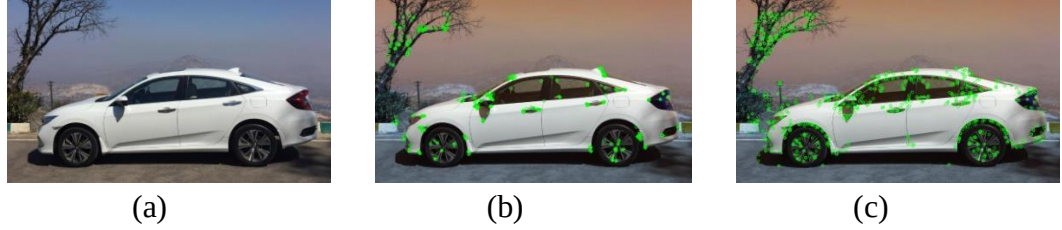


Figure 3.7: Example 1 - (a) original image , (b) ORB key points , (c) AKAZE key points

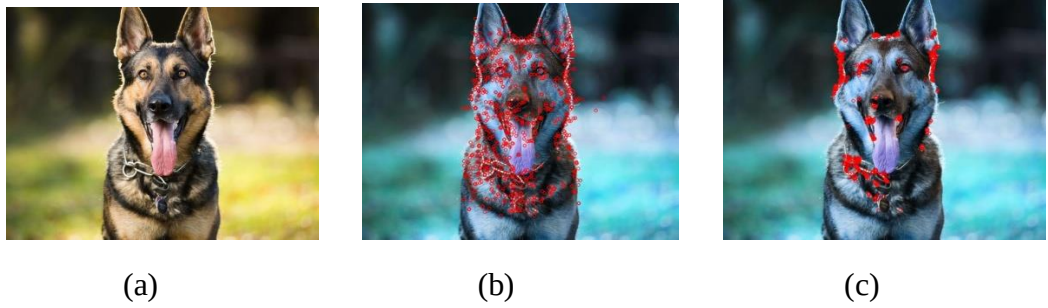


Figure 3.8: Example 2 - (a) original image , (b) ORB key points , (c) AKAZE key points

3.1.3 Moving object key points Detection using Energy function

The energy is a measure of the localized change of the image. It's the rate of change in the color/brightness/magnitude of the pixels over local areas[32]. To detect the moving object key points or pixels in each frame Energy function is used by calculating Entropy.

Entropy is the measurement of the dissimilarity of the pixels [29]. In this method entropy defines that the key point belongs to foreground or background.

Entropy is formulated as follows:

$$-\sum p(x) * \log p(x) \quad \dots(3.11)$$

Where $p(x)$ is the probability of a given symbol/event.

n number of frames extracted from the input video, then key points are extracted from all the frames. But to calculate the energy of pixels in each frame at a time 3 frames are taken and that is act like a Bucket. Moving object detection method will applied to each bucket individually. Suppose there are 30 total frames, then $30/3 = 10$ times this method is applied.

From n number of frames 3 frames are taken each time is shown in figure 3.9 below:

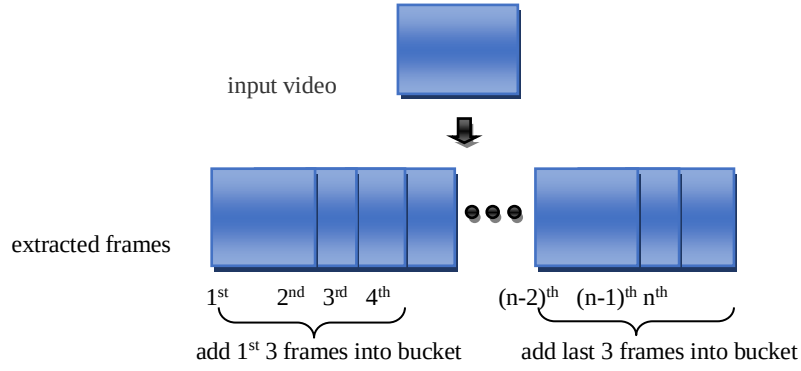


Figure 3.9: Addition of 3 frames into one bucket before applying detection method

Suppose the first bucket is taken where first 3 continuous frames are $f_1(x,y)$, $f_2(x,y)$, $f_3(x,y)$ respectively [31].

Let, $f_1(x,y)$ has a numbers of key points ,

$$kp_1^{total} = \{ k_{11}, k_{12}, k_{13}, \dots, k_{1a} \} \quad \dots(3.12)$$

The second frame $f_2(x,y)$ has b numbers of key points ,

$$kp_2^{total} = \{ k_{21}, k_{22}, k_{23}, \dots, k_{2b} \} \quad \dots(3.13)$$

And the last frame of the bucket $f_3(x,y)$ has c numbers of key points ,

$$kp_3^{total} = \{ k_{31}, k_{32}, k_{33}, \dots, k_{3c} \} \quad \dots(3.14)$$

All key points from these 3 key points sets are taken by doing union among those sets because if one key point is redundant then only once it is taken and if one key point is not present in one frame but not in other two then it is also considered.

$$K_{123} = (kp_1^{total} \cup kp_2^{total} \cup kp_3^{total}) \quad \dots(3.15)$$

Now from this K_{123} every pixels are taken one by one.

Suppose m^{th} pixel $p(x_m, y_m)$ is a key point in K_{123} taken from the 3 frames. Now intensities of the pixel in each frame are $I_1(x_m, y_m)$, $I_2(x_m, y_m)$, $I_3(x_m, y_m)$ respectively. Pictorial representation of the intensities of the same pixel co-ordinate is shown below in Figure 3.10.

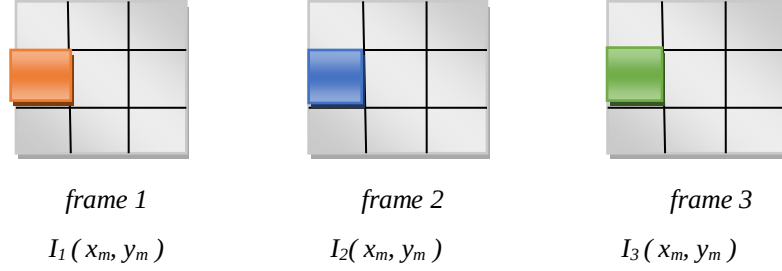


Figure 3.10: Intensity value of the same pixels co-ordinates from 3 consecutive frames

Mean of 3 intensities $I_1(x_m, y_m)$, $I_2(x_m, y_m)$, $I_3(x_m, y_m)$ is calculated :

$$\mu_m = \frac{1}{n} \sum_{i=1}^n I_i(x_m, y_m) \quad \text{where } (n = 3) \quad \dots(3.16)$$

Where n is the total number of frames.

Standard deviation of those intensities of m^{th} pixel $p(x_m, y_m)$ is calculated as :

$$\sigma_m = \sqrt{\left(\frac{1}{n-1}\right) \sum_{i=1}^n (I_i(x_m, y_m) - \mu_m)^2} \quad \text{where } (n = 3) \quad \dots(3.17)$$

Where n is the total number of frames, μ_m is mean (3.16).

Now to determine the taken pixel $p(x_m, y_m)$ is a part of the static background or part of any moving object, entropy is calculated and that entropy value is assigned to the particular pixel. entropy is calculated using Gaussian Distribution / Normal Distribution [30] which is used.

In the first frame $f_1(x, y)$ for the m^{th} pixel $p_1(x_m, y_m)$, the Gaussian/normal distribution G_1 is calculate as[27]:

$$G_1 = \frac{1}{\sqrt{2\pi\sigma_m^2}} e^{-\frac{1}{2} \left(\frac{I_1(x_m, y_m) - \mu_m}{\sigma_m} \right)^2} \quad \dots(3.18)$$

Where μ_m is mean using (3.16) , σ_m is standard deviation using (3.17) , $I_1(x_m, y_m)$ is the intensity of m^{th} pixel $p_1(x_m, y_m)$ in the first frame $f_1(x, y)$.

Likewise in the bucket for the rest 2 frames $f_2(x, y)$, $f_3(x, y)$ Gaussian distribution G_2 , G_3 are calculated for the same m^{th} pixel $p_2(x_m, y_m)$, $p_3(x_m, y_m)$.

The final entropy for $p(x_m, y_m)$ (for the three consecutive frame) using Shannon Entropy (3.11) is calculated and denoted as E_m :

$$E_m = \sum_{i=1}^n G_i * \log (G_i) \quad \text{where } (n = 3) \quad \dots(3.19)$$

Likewise Entropy is calculated for all key points / pixel in K_{123} (for three frames $f_1(x, y)$, $f_2(x, y)$, $f_3(x, y)$) as m^{th} pixel which is also a keypoint in K_{123}). Suppose there are v no of key points in K_{123} and kept in E set followed as :

$$E = \{E_1, E_2, E_3, \dots \dots \dots, E_v\} \quad \dots(3.20)$$

This is important step where previously calculated E_m is filtered with a threshold, which decides the taken pixel m^{th} pixel is a part of moving object or static background.

The threshold is Mean of all entropies in set E divided by square of total number of frames are detected in one bucket. Threshold t is calculated as :

$$t = \left(\frac{1}{v} \sum_{i=1}^v E_i \right) / n^2 \quad \text{where } (n = 3) \text{ and } (E_i \in E) \quad \dots(3.21)$$

Now the entropy of m^{th} pixel E_m is compared with the threshold t . If E_m is less than t then the pixel $p(x_m, y_m)$ comes under the dynamic foreground, else it belongs to the static background.

$$if(E_m < t) \begin{cases} 1, & p(x_m, y_m) \in foreground \\ 0, & p(x_m, y_m) \in background \end{cases} \quad \dots(3.22)$$

Likewise every entropy in set E is filtered by this threshold t and in the same way all the foreground key points are detected. All the pixels belongs to moving objects separated from those pixel those belong to static background and kept in $K_{foreground}$ set as follows:

$$K_{foreground} = \{k_1, k_2, k_3, \dots \dots \dots\} \quad \text{where } (k_i \in foreground) \quad \dots(3.23)$$

Where k_i belongs to foreground pixels for 3 frames $f_1(x,y)$, $f_2(x,y)$, $f_3(x,y)$. So to get moving objects of first frame $f_1(x,y)$, all key points extracted for the frame which was stored in kp_1^{total} (3.12) are intersected with the foreground key points for 3 frames stored in $K_{foreground}$. After intersection the key points are kept in $kp_1^{foreground}$ which are moving object key points / pixels in the first frame.

$$kp_1^{foreground} = (kp_1^{total} \cap K_{foreground}) \quad \dots(3.24)$$

Similarly, $kp_2^{foreground}$, $kp_3^{foreground}$ are extracted for $f_2(x,y)$, $f_3(x,y)$:

$$kp_2^{foreground} = (kp_2^{total} \cap K_{foreground}) \quad \dots(3.25)$$

$$kp_3^{foreground} = (kp_3^{total} \cap K_{foreground}) \quad \dots(3.26)$$

3.1.4 Grouping foreground key points into individual objects

After getting the foreground key points of each frame main task is to group those individual objects and this task is to be done with the help of foreground key points only. It's observed that key points of an objects are close to each other or more concentrated than rest of the places where moving object is not present. Based on this observation object would get a proper definite shape in the frame by tweaking the pixels/key points with the help of morphological operation. Basically two morphological operations [33] are used here Dilation and Erosion.

Now first frame $f_1(x,y)$ is taken and it is transformed into binary image. where those key points / pixels from $kp_1^{foreground}$ (foreground key points) are white and rest pixels are black in color.

So take each pixel of that frame $f_1(x,y)$ and suppose that frame has total Q no of pixels,

$$if (p(x_m, y_m) \in kp_1^{foreground}) = \begin{cases} 1, & p(x_m, y_m) = 1(white) \\ 0, & p(x_m, y_m) = 0(black) \end{cases} \quad \dots(3.27)$$

where $m = \{1, 2, \dots, Q\}$ and after image binarization the frame $f_1(x,y)$ is now denoted as B_1 .

At first to grow the size of the pixels dilation is done. This procedure follows convolution with 5×5 square kernel k_1 containing all 1's. This kernel has an anchor point, which is middle pixel (3,3) of the kernel k_1 . The dilation is formulated as [34]:

$$B_1 \oplus k_1 = z \mid (k_1 \cap B_1) \in B_1 \quad \dots(3.28)$$

When the image B_1 is dilated by the structuring element/ kernel k_I , the outcome element z would be that there will be at least one element in k_I , that intersects with an element in B_1 . That means if at least one of pixel in the kernel k_I matched with the pixel of B_1 , situated in the same position within the kernel window then the center pixel B_1 in is replaced with 1 else 0.

Then slide the kernel to the right and by doing this move downwards of the image. With this procedure, the areas of white regions grow in size and hence the object size increases. The process of dilation given below in the Figure 3.11.

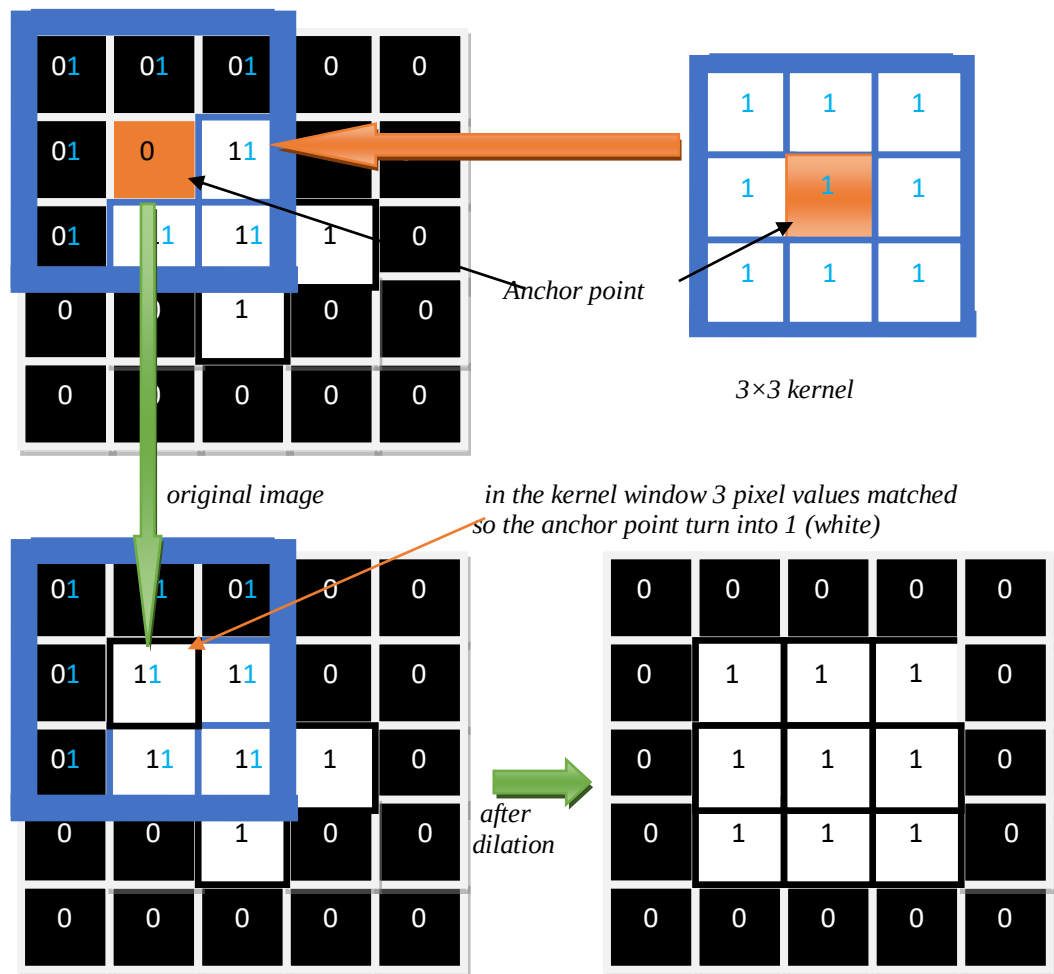


Figure 3.11 : The process of Dilation

Due to dilation the pixels are expanded to make them larger and prominent several times dilation is done and now the image is denoted as $B_1^{dilated}$. But there exists some noisy key points which are not part of moving objects, also expanded during dilation, to remove them or to make less prominent few times Erosion is done.

The erosion is done on $B_1^{dilated}$ and This procedure has the same 5×5 square kernel k_1 with all 1's. As before this kernel has an anchor point, which is middle pixel (3,3) of the kernel k_1 . The dilation is formulated as :

$$B_1^{dilated} \ominus k_1 = z \mid (k_1 \in B_1^{dilated}) \quad \dots(3.29)$$

When the image $B_1^{dilated}$ is dilated by the structuring element/ kernel k_1 , the outcome element z is considered only when the kernel k_1 is a subset or equal to the binary image $B_1^{dilated}$.

That means if all the pixels in the kernel k_1 matched with all the pixel of B_1 situated in the same position within the kernel window then the center pixel B_1 in is replaced with 1 else replaced with 0.

Then similarly slide the kernel to the right and by doing this move downwards of the image. With this procedure, the areas of white regions are reduced in size and hence the object sizes are decreased[35]. The process of erosion given below in the Figure 3.12.

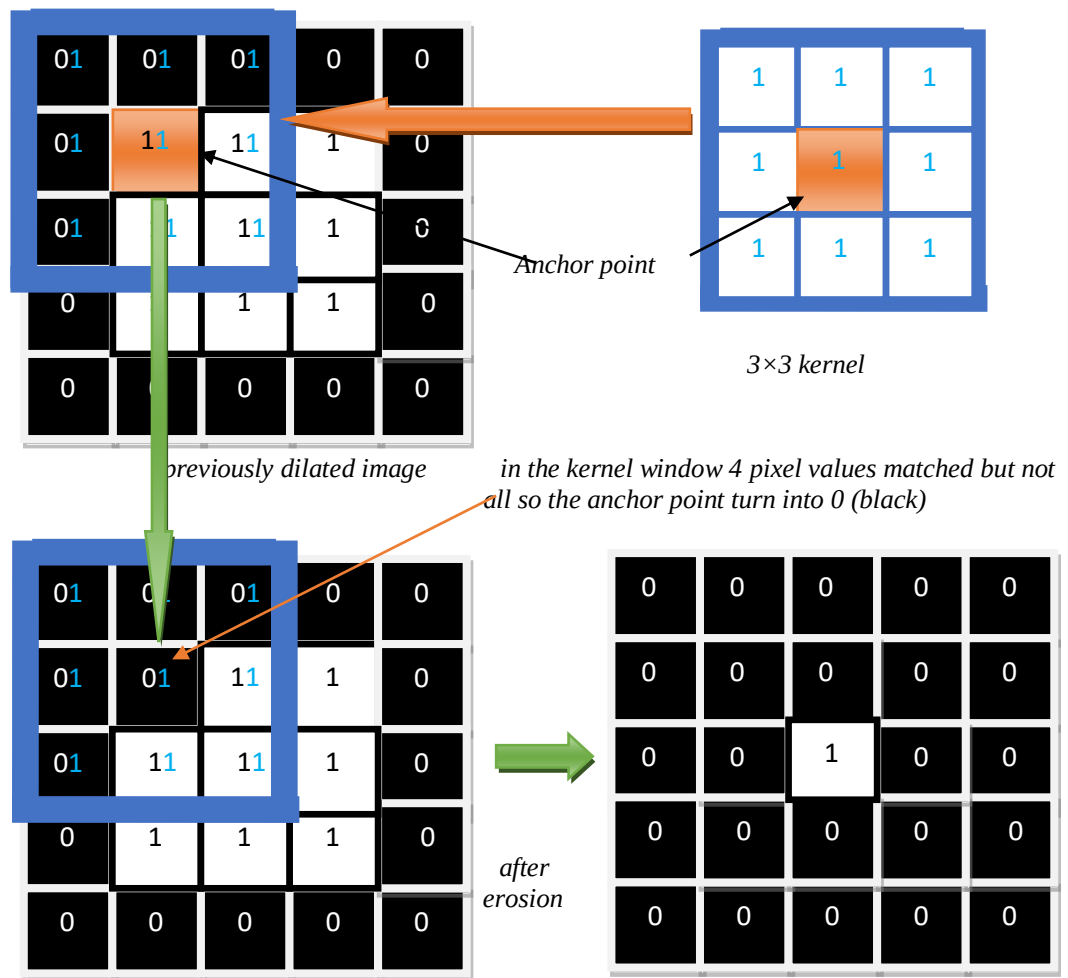


Figure 3.12: The process of Erosion

After dilation and erosion a rough structures of the moving objects are extracted and that is stored in $B_I^{dilated_eroded}$.

But to draw the bounding boxes around the objects, actual locations of those objects are needed, for that blob detection[36] method is used. Here blob detection method is used to group the white pixels and it detects the white pixels based on the intensity of the white pixel and the 8 connectivity[37].

Blob detection method is applied on the binary image $B_I^{dilated_eroded}$, where the white pixels are the object pixels and black pixels are the back ground. So blobs are detected around the white pixels.

If the m^{th} pixel $p_1(x_m, y_m)$ of $B_I^{dilated_eroded}$ then the intensity of the pixel would be $I_1(x_m, y_m)$.

Suppose the first blob is $blob_1$ where the white pixels are stored based on 8-connectivity and each pixel in the image will be checked from the top left position. the conditions are for blob detection are :

- if $((I(x_m, y_m) = 0(\text{white})))$ and it is the first white pixel, then include $p(x_m, y_m)$ to $blob_1$.
- if $((I(x_m, y_m) = 0(\text{white})))$ and it is not the first white pixel and at least one of the 8 neighbor pixels of $p(x_m, y_m)$ are already present in $blob_1$ then include $p(x_m, y_m)$ to $blob_1$.
- else if $((I(x_m, y_m) = 1(\text{black})))$ then just move to next pixel.

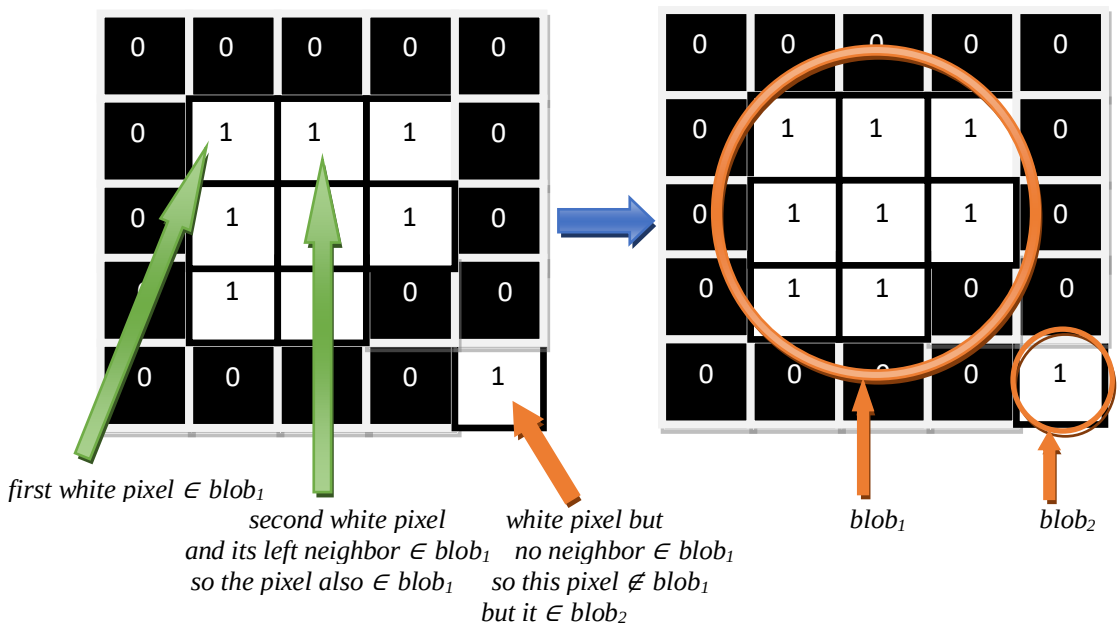


Figure 3.13 : Blob detection procedure

In the above Figure 3.13 we can see 2 blobs has been detected. Let bigger blob $blob_1$ is the actual object but the smaller one $blob_2$ is basically considered as noise. So by putting the threshold of the area the smaller blobs are avoided as those are considered as non-object blob or noise points.

By this blob detection method center and diameter of each blob are detected. With this information bounding boxes are drawn.

Let the center of a blob is $c_1(x_i, y_i)$ and the diameter is d_1 so the radius r_1 is $d_1/2$. So the dimension of the bounding box would be and the pictorial representation is in 3.14 and 3.15 –

Top left corner = $(x_i - r_1, y_i + r_1)$
 Top right corner = $(x_i + r_1, y_i + r_1)$
 Bottom left corner = $(x_i - r_1, y_i - r_1)$
 Bottom right corner = $(x_i + r_1, y_i - r_1)$

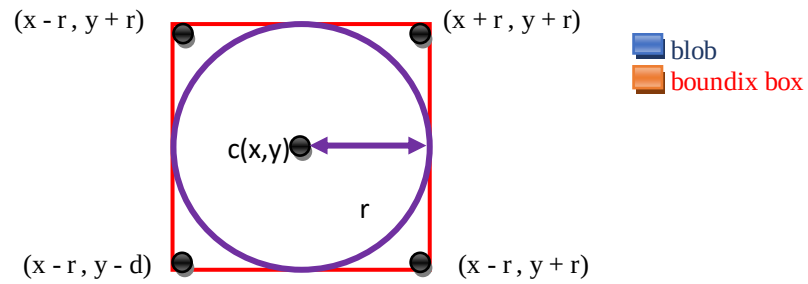


Figure 3.14 : Dimensions of bounding box calculated from blob center and diameter

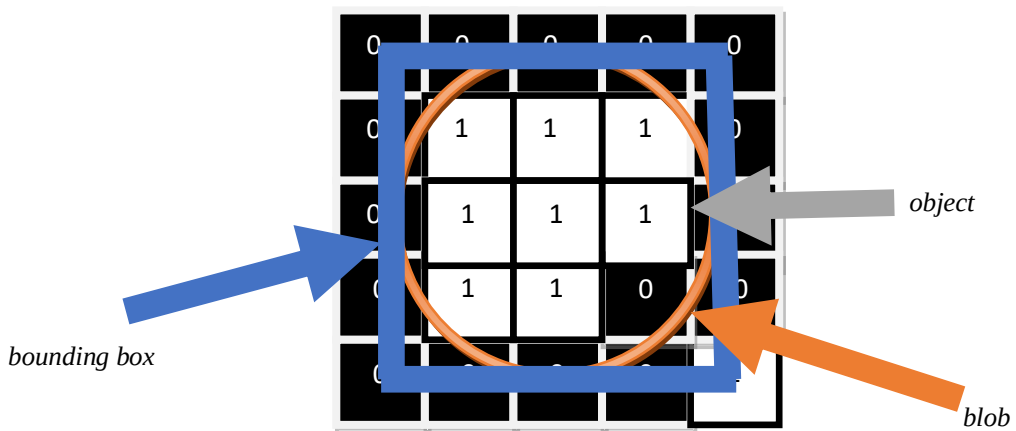


Figure 3.15: Pictorial representation of bounding box around the object

Chapter 4:

Experimental Results

This section demonstrates some of the tested image sequences that are able to highlight the effectiveness of each steps of the proposed moving object detection system. These sequences used in experiments consist of indoors and outdoors testing environments so that the proposed scheme can be fully evaluated.

Testing of the proposal method is done on two videos – one indoor and outdoor videos. First key points / interesting points are detected by ORB and KAZE methods in a frame. Figure 4.1 shows the key points in a frame of an indoor and outdoor video.

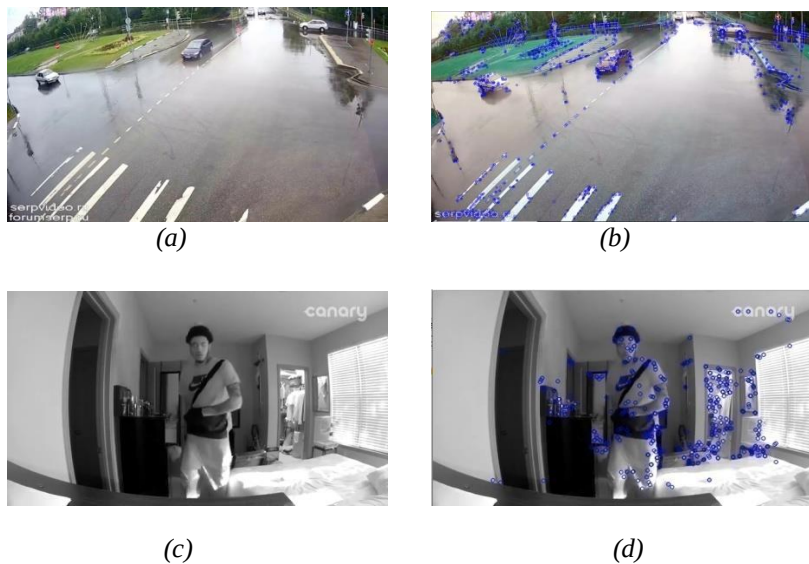


Figure 4.1: (a),(c) Original image (b),(d) extracted key points by ORB and KAZE

Then using energy function entropy is assigned to those key points and by applying threshold moving objects are detected. Figure 4.2 shows moving objects key points in indoor and outdoor frames.

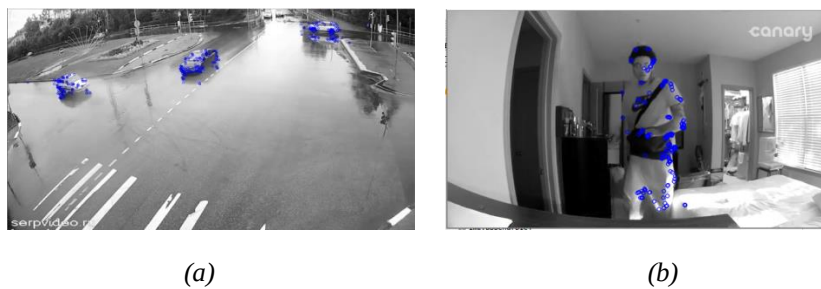


Figure 4.2:(a) Outdoor (b) indoor moving object key points by energy function and filtered by threshold

Then using this key points morphological operations done after turning it into black and white image. Enlarging the key points (white pixels) for getting a proper shape to localize the moving objects. With the help of blob detector circles are drawn around those white enlarged objects (white pixels) and diameters and centers of the circles are detected. With those measurements bounding boxes are drawn around the objects.

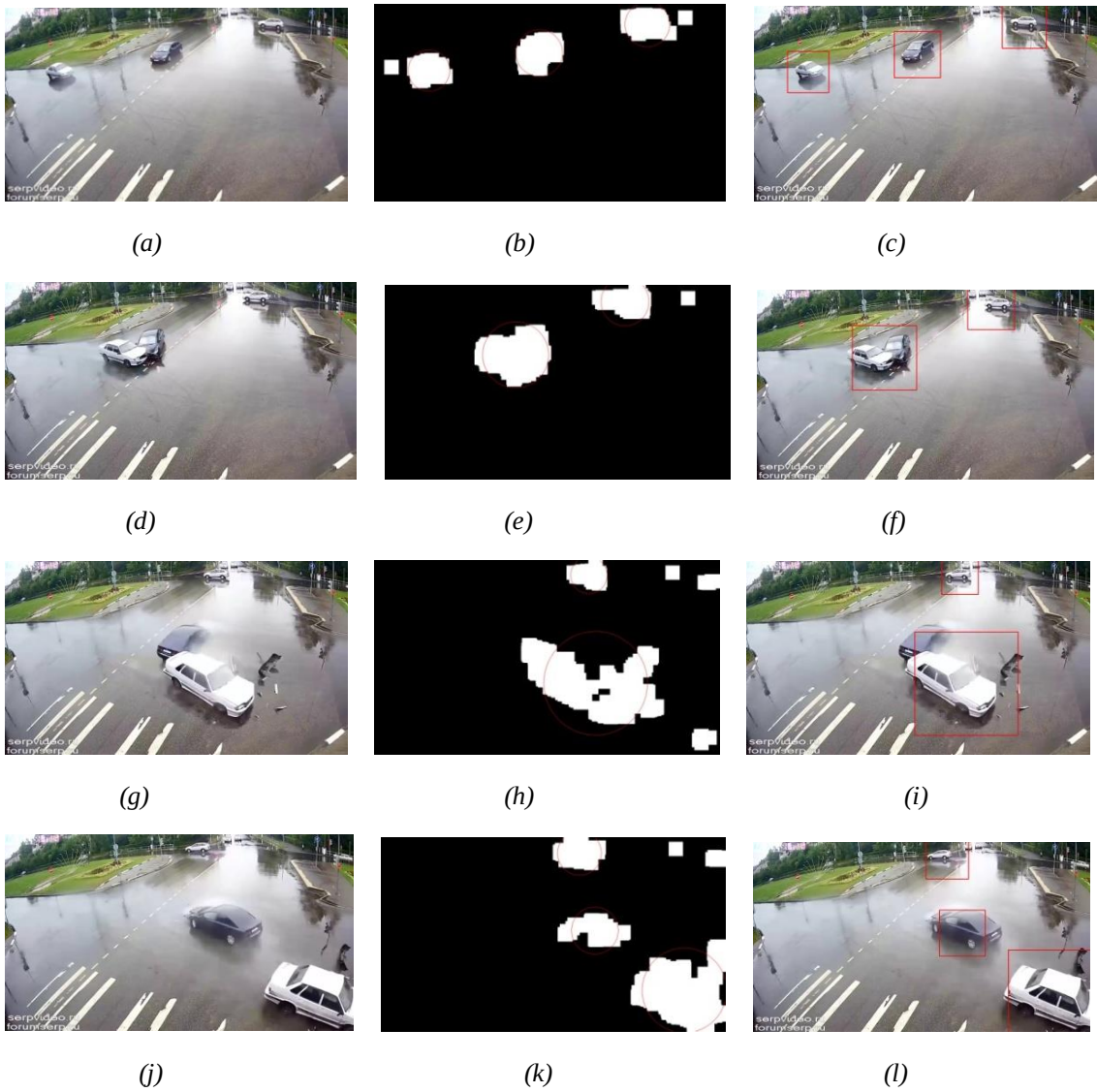


Figure 4.3 : Outdoor video results : (a),(d),(g),(j) original frames, (b),(e),(h),(k) subtraction of moving objects (c),(f),(i),(l) moving objects with bounding boxes.

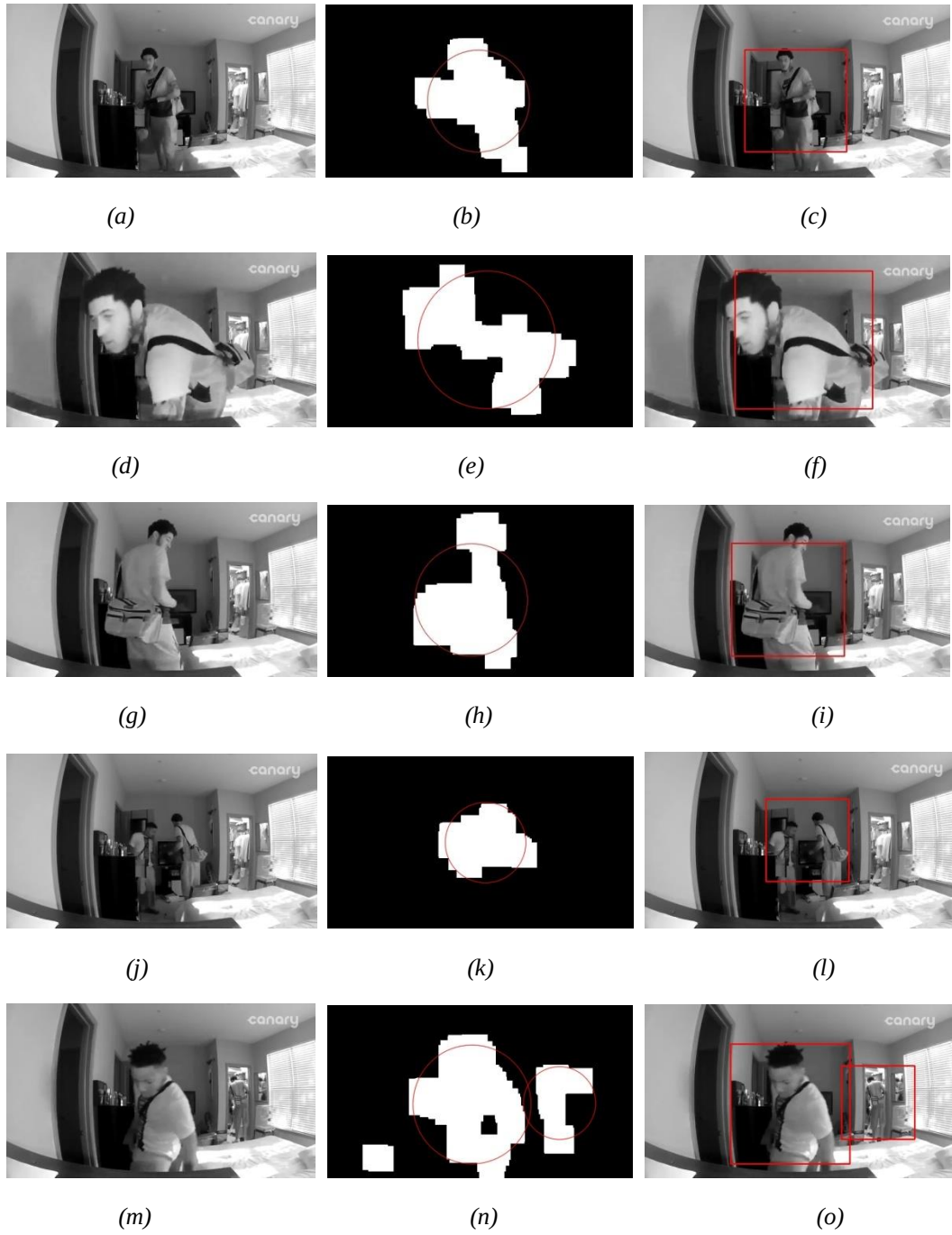


Figure 4.4 : Indoor video results : (a),(d),(g),(j),(m) original frames , (b),(e),(h),(k),(n) subtraction of moving objects, (c),(f),(i),(l),(o) moving objects with bounding boxes.

In the above results of testing of the proposed moving object detection method is shown. This method is very simple and has very less time complexity as it does not use very high complex algorithm.

This method has two shortcomings ,

- One it works only on with static background. if the background also dynamic it could not able to detect, as it cannot differentiate between foreground and background. Mostly it will consider the whole frame as one moving object.
- Another one is it cannot work properly on those video having highly congested moving objects. That means too many moving objects are placed at very close places. It will consider those objects as a single object with single boundary box so the accuracy will drastically decrease.

Though having these short comings this method gives descent detection result with simple indoor outdoor videos which recorded through a static camera.

YouTube link for the experimental result : <https://youtu.be/UD69kYWT2n4>

Chapter 5:

Conclusion and Future Scope

Object detection and tracking is an important task in computer vision field. Object detection in video image obtained from single camera with static background that means the camera is fixed at a location and detection is achieved by background subtraction approach. In this thesis, we tried different videos with fixed camera with a single object and multiple objects to see it is able to detect objects.

The algorithm is able to detect and track a moving object that is moving. The input for this project is video sequences which were captured via traffic camera (static) to detect moving cars /vehicles and CCTV camera installed for indoor to detect human activity. The first step of the algorithm is to sample the video sequence into static frames. This train of frames is originally in red-green-blue (RGB). To ease computation, then RGB frames are converted to Grey scale. Then entropy value is put to each frame, subtraction of the moving object key points is done by filtering with a threshold. Finally by modifying the frames with morphology and blob detection moving object is detected and bounded by a rectangular bounding box. Moving Object detection segments the moving targets from the background and it is the crucial first step in video surveillance. This algorithm is applied on indoor and as well as outdoor videos and successfully detected the moving objects. This method is less complex and may not highly accurate but it gives descent result within less time.

This method has two shortcomings, one it works only on with static background. If the background also dynamic it could not able to detect, as it cannot differentiate between foreground and background. Mostly it will consider the whole frame as one moving object. Another one is it cannot work properly on those video, which has highly congested moving objects, means too many moving objects at very short places. It will consider objects as a single object with single boundary box so the accuracy will drastically decrease.

In the future, we can extend the work to detect the moving object with non-static background, having multiple cameras which can be used in real time surveillance applications and faster and more suitable for surveillance applications where congestion is very high. Then it will give the most promising results in terms of detection quality, speed and computational complexity.

Chapter 6:

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